SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Department of Artificial Intelligence and Machine Learning

Data Visualization Lab — Practical No. 6

DAX Measures, Time-Based Analysis and Interactive Dashboard Using Power BI

Course: Data Visualization Lab	**Course Code:** TE7663
Academic Year: 2026-2027	**Semester:** V
Student Name: **Neel Lakhani**	**PRN / Roll No.:** **24070126120**

Blog Link:

1. Aim and Objectives

Aim

The aim of this practical is to use Microsoft Power BI and DAX Query View to analyze the AdventureWorks dataset, create important business performance indicators, perform time-based calculations, and develop an interactive sales dashboard.

Objectives

The main objectives of this practical are:

- To understand and prepare the AdventureWorks dataset for analysis.
- To create important KPI measures such as Total Sales, Total Profit, Total Orders, Total Customers, Average Order Value and Profit Margin.
- To create at least 20 custom DAX measures using different DAX functions.
- To use DAX Query View to execute queries and validate calculations.
- To develop an interactive Power BI dashboard containing KPI cards, charts, a matrix, a scatter plot and slicers.

2. Dataset Description and Tools Used

2.1 Dataset Description

For this practical, I used the AdventureWorks dataset to analyse sales performance using Power BI. The dataset contains information about sales transactions, products, customers, sales territories, and dates. These details help in understanding sales trends, comparing product categories, and analysing customer purchasing patterns.

The dataset consists of the following tables:

Table Name	Description
Sales Data	Contains sales records, order quantities and related sales information.
Product Lookup	Contains product names, colors, prices, costs and product identifiers.
Product Subcategory and Category	Helps compare sales across product groups.
Customer Data	Provides customer information and supports unique-customer calculations.
Territory Data	Helps analyze sales by country, region and continent.
Calender	Supports monthly, yearly and previous-period sales calculations.

Each table serves a different purpose and provides information required for sales analysis. The Sales table acts as the main transaction table, while the remaining tables provide additional details about products, customers, territories, and dates.

2.2 Tools Used

The following tools were used for this practical:

Microsoft Power BI Desktop: Used to import data, create relationships, develop DAX measures and design the dashboard.

Power Query: Used to prepare and organize the data before analysis.

DAX (Data Analysis Expressions): Used to calculate KPIs, sales growth, profit margin and other business measures.

DAX Query View: Used to write and execute DAX queries within Power BI Desktop.

Microsoft Word: Used to prepare the practical report.

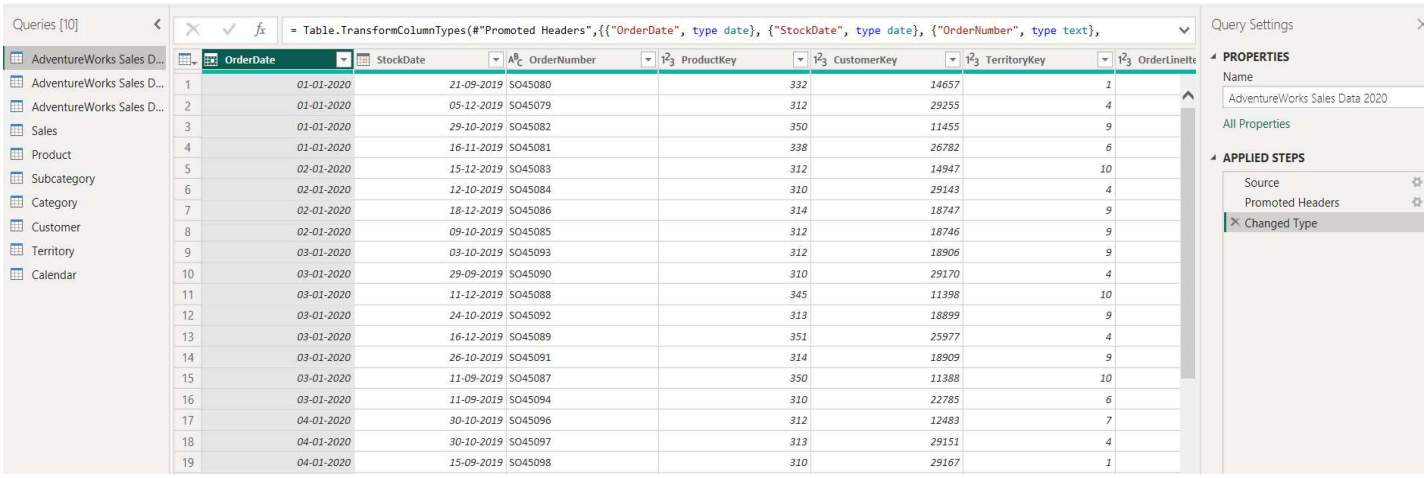
3. Data Preparation and Relationships

3.1 Data Preparation

The first step was to import the AdventureWorks dataset into Power BI Desktop. The required tables, including Sales, Product, Subcategory, Category, Customer, Territory, and Calendar, were loaded into Power Query Editor.

After importing the data, I checked the column names, data types, and overall structure of each table. Date columns were assigned the Date data type, quantity columns were set to Whole Number, and price and cost columns were set to Decimal Number. I also checked that the key columns had matching data types so that the tables could be connected correctly.

The data was then loaded into Power BI using the Close & Apply option.



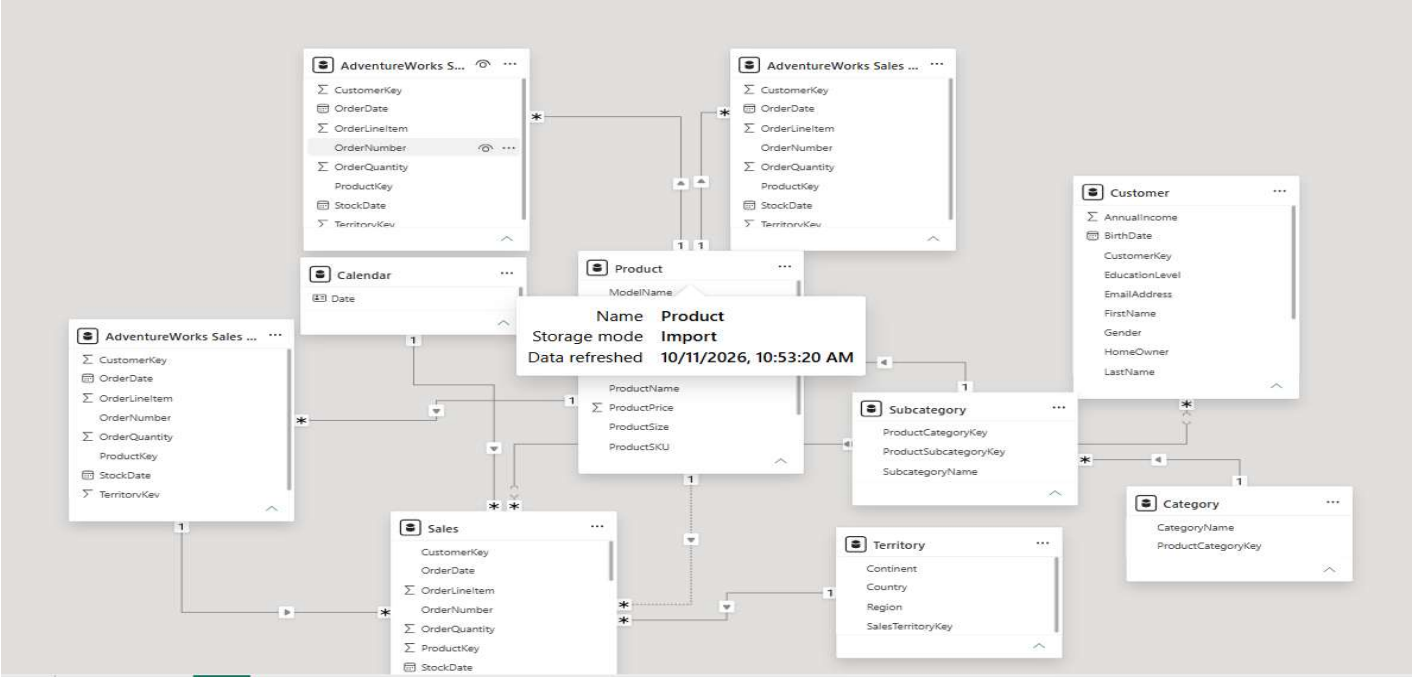
2.2 Creating Relationships Between Tables

After preparing the data, I moved to Model view in Power BI to create relationships between the tables. These relationships connect the sales data with the corresponding product, customer, territory, and calendar information.

The Sales table acts as the main transaction table, while Product, Customer, Territory, and Calendar provide additional information for analysis. The Product table is also connected to Subcategory, which is connected to Category.

The following relationships were created:

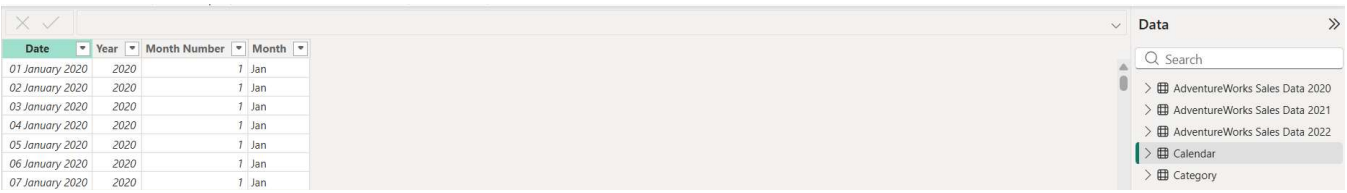
One side (1)	Many side (*)	Columns
Calendar	Sales	Date → OrderDate
Product Sales	Sales	ProductKey → ProductKey
Customer	Sales	CustomerKey → CustomerKey
Territory	Sales	SalesTerritoryKey → TerritoryKey
Category	Subcategory	ProductCategoryKey → ProductCategoryKey
Subcategory	Product	ProductSubcategoryKey → ProductSubcategoryKey



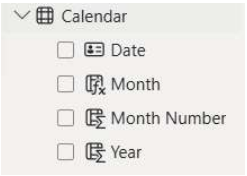
2.3 Calendar Table Preparation

The Calendar table was marked as a date table using the Calendar[Date] column. Three calculated columns—Year, Month Number, and Month—were added using DAX.

The Month column was sorted by Month Number to display the months in the correct calendar order. These columns help analyse sales by year and month and make time-based visualisations easier to create.



Date	Year	Month Number	Month
01 January 2020	2020	1	Jan
02 January 2020	2020	1	Jan
03 January 2020	2020	1	Jan
04 January 2020	2020	1	Jan
05 January 2020	2020	1	Jan
06 January 2020	2020	1	Jan
07 January 2020	2020	1	Jan



3 DAX Measures and KPI Calculations

DAX measures were used to calculate important sales and profitability figures. Measures are useful because their results can change automatically when a user selects a different year, region or other filter in the dashboard.

The following formulas are examples for the AdventureWorks model. The table and column names should be adjusted to match the actual model in Power BI.

3.1 Core KPI Measures

1. Total Sales

Formula :

```
Total Sales =
SUMX(
    'Sales',
    'Sales'[OrderQuantity] *
    RELATED(Product[ProductPrice])
)
```

Explanation: This calculates sales by multiplying the quantity ordered by the related product price and adding the results together.





2. Total Cost

```
Total Cost =
SUMX(
    'Sales',
    'Sales'[OrderQuantity] *
    RELATED(Product[ProductCost])
)
```

Explanation: This calculates the total product cost using the quantity ordered and the corresponding product cost.

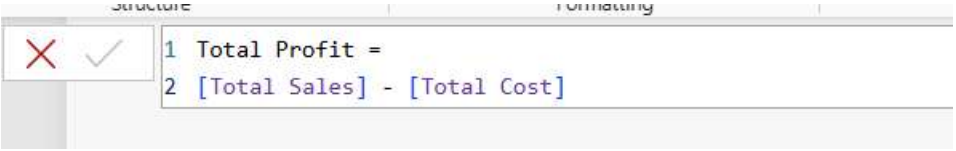


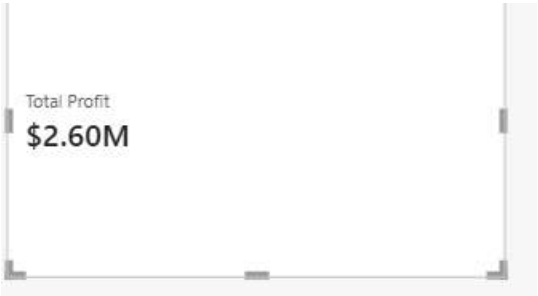


3. Total Profit

```
Total Profit =
[Total Sales] - [Total Cost]
```

Explanation: This calculates the difference between total sales and total product cost.

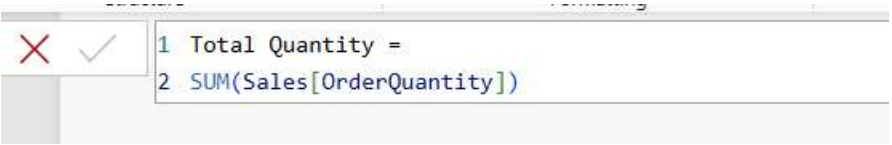


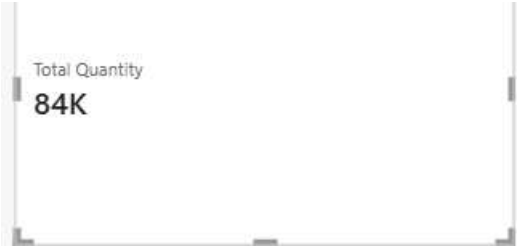


4. Total Quantity

```
Total Quantity =
SUM('Sales'[OrderQuantity])
```

Explanation: This measure calculates the total number of units ordered.



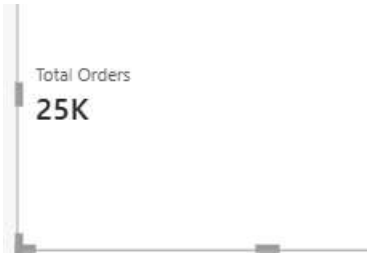


5. Total Orders

```
Total Orders =
DISTINCTCOUNT('Sales'[OrderNumber])
```

Explanation: This counts unique order numbers instead of counting every individual order line.

```
Total Orders =
DISTINCTCOUNT(Sales[OrderNumber])
```



6. Total Customers

```
Total Customers =
DISTINCTCOUNT('Sales'[CustomerKey])
```

Explanation: This counts unique customers represented in the sales data.

```
Total Customers =
DISTINCTCOUNT(Sales[CustomerKey])
```



7. Average Sales

```
Average Sales =
AVERAGE('Sales'[Sales Amount])
```

Explanation: This calculates the average sales amount per row. It requires a valid sales amount column.

```
Average Sales =
AVERAGEX(
    Sales,
    Sales[OrderQuantity] *
    RELATED(Product[ProductPrice])
)
```

Average Sales
$2.435335962091K

8. Average Order Value

```
Average Order Value =
DIVIDE(
    [Total Sales],
    [Total Orders],
    0
)
```

Explanation: This calculates the average sales value for each unique order.

```
Average Order Value =
DIVIDE([Total Sales], [Total Orders], 0)
```

Average Order Value
$254.527641881252

9. Average Product Price

```
Average Product Price =
AVERAGE(Product[ProductPrice])
```

Explanation: This calculates the average listed price of the products.

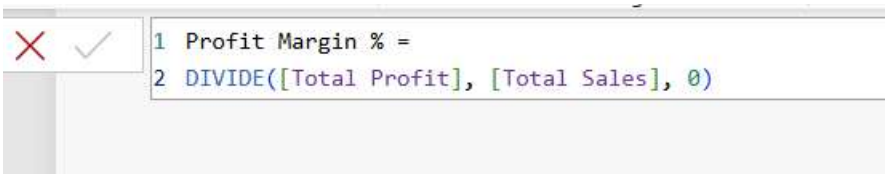
```
Average Product Price =
AVERAGE(Product[ProductPrice])
```

Average Product Price
$714.4373566552902

10. Profit Margin Percentage

```
Profit Margin % =
DIVIDE(
    [Total Profit],
    [Total Sales],
    0
)
```

Explanation: This calculates profit as a proportion of sales. The measure should be formatted as a percentage in Power BI.





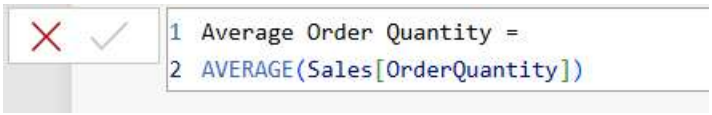
3.2 Additional Custom DAX Measures

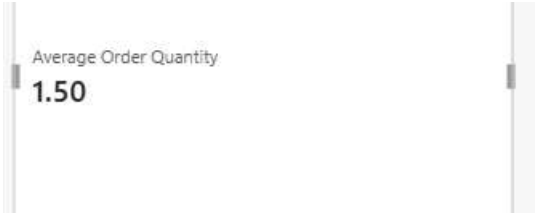
The assignment also requires at least 20 custom DAX measures. The following examples extend the basic KPIs to cover additional calculations.

11. Average Quantity per Order

```
Average Quantity per Order =
DIVIDE(
    [Total Quantity],
    [Total Orders],
    0
)
```

This calculates the average number of units per unique order.

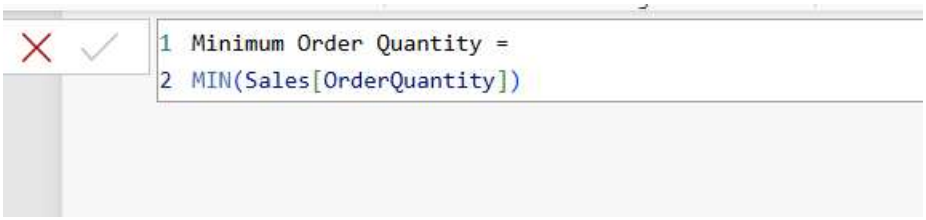


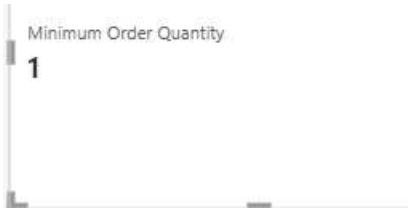


12. Minimum Product Price

```
Minimum Product Price =
MIN(Product[ProductPrice])
```

This identifies the lowest listed product price.

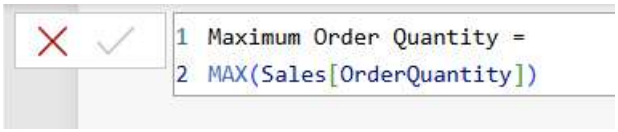




13. Maximum Product Price

```
Maximum Product Price =
MAX(Product[ProductPrice])
```

This identifies the highest listed product price.

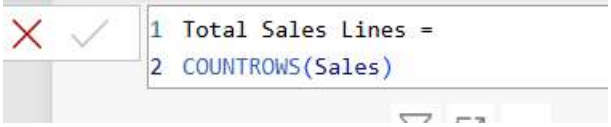




14. Total Sales Line

```
Total Sales Lines =
COUNTROWS(Sales)
```

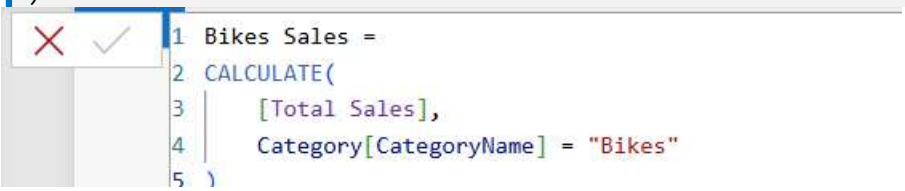
Counts the transaction lines in the sales table.





15. Sales From Bikes

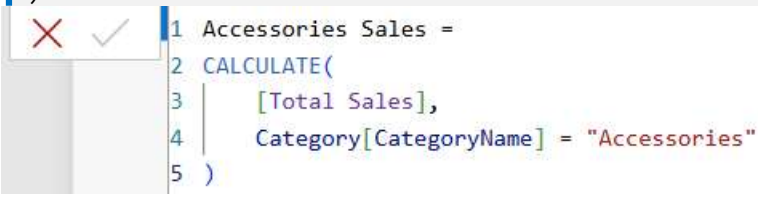
```
Bikes Sales =
CALCULATE(
    [Total Sales],
    Category[CategoryName] = "Bikes"
)
```





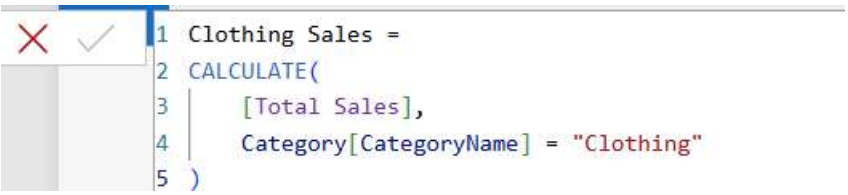
16. Sales From Accessories

```
Accessories Sales =
CALCULATE(
    [Total Sales],
    Category[CategoryName] = "Accessories"
)
```



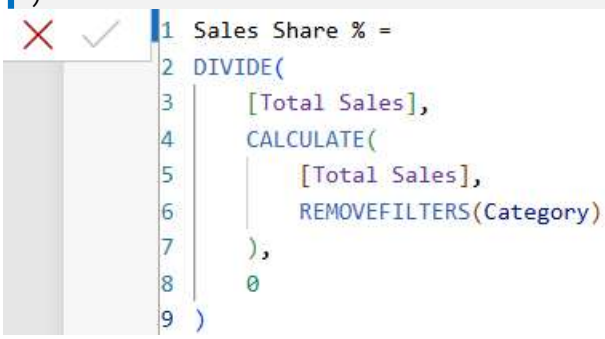
17. Sales From Clothing

```
Clothing Sales =
CALCULATE(
    [Total Sales],
    Category[CategoryName] = "Clothing"
)
```



18. Sales Shares %

```
Sales Share % =
DIVIDE(
    [Total Sales],
    CALCULATE(
        [Total Sales],
        REMOVEFILTERS(Category)
    ),
    0
)
```

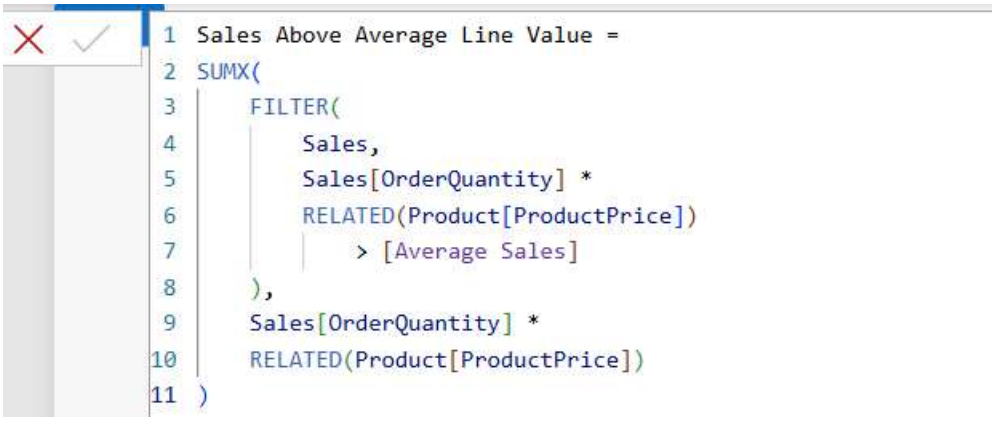




Uses FILTER to sum sales lines whose value exceeds the average line value. This is a more advanced measure

19. Sales Above Average Line Value

Sales Above Average Line Value = SUMX(FILTER(Sales, Sales[OrderQuantity] * RELATED(Product[ProductPrice]) > [Average Sales]), Sales[OrderQuantity] * RELATED(Product[ProductPrice]))

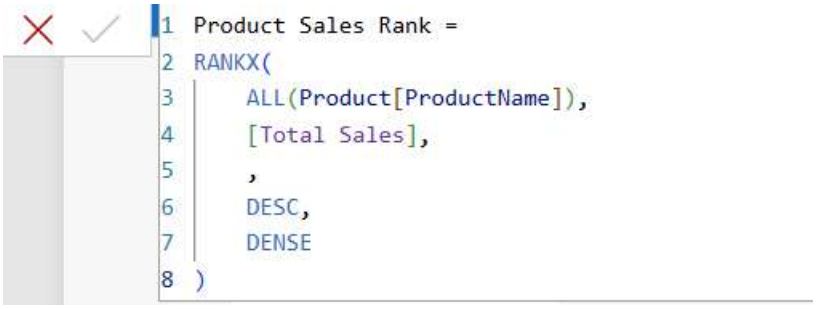


20. Product Sales Rank

```
Product Sales Rank =
RANKX(
    ALL(Product[ProductName]),
    [Total Sales],
    ,
    DESC,
    DENSE
)
```

Product Sales Rank

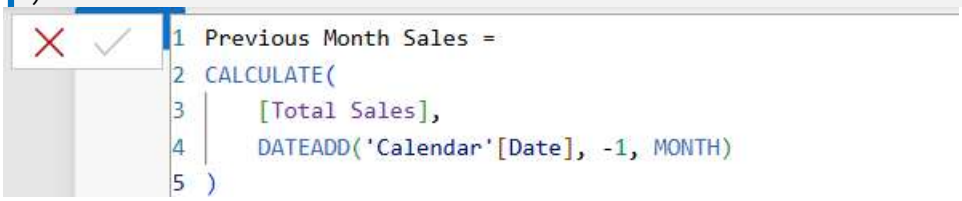
1



Rank 1 indicates the highest-selling product in the current context.

21. Previous Month Sales

```
Previous Month Sales =
CALCULATE(
    [Total Sales],
    DATEADD('Calendar'[Date], -1, MONTH)
)
```

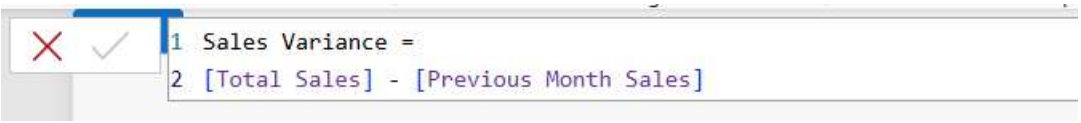


Previous Month Sales

6.40M

22. Sales Variance

```
Sales Variance =
[Total Sales] - [Previous Month Sales]
```





3.3 Importance of DAX Measures

DAX measures help convert raw records into useful business information. For example, Total Sales gives an overview of sales performance, Total Profit helps assess profitability, and Profit Margin shows how much profit is generated relative to sales.

Time-based measures make it possible to identify increases or decreases over time, while ranking measures help identify the products contributing the most sales.

4. DAX Query View

DAX Query View is useful for testing calculations and checking whether the measures return the expected results.

DAX Query View is available within Power BI Desktop. DAX Studio can also be connect to an open Power BI model and execute queries against it.

For this practical, queries can be used to compare the KPI results with the values displayed on the dashboard.

4.1 Query to Validate the Main KPIs

The following query returns the core KPI measures in one row.

```
EVALUATE
ROW(
    "Total Sales", [Total Sales],
    "Total Cost", [Total Cost],
    "Total Profit", [Total Profit],
    "Total Orders", [Total Orders],
    "Total Customers", [Total Customers],
    "Profit Margin %", [Profit Margin %]
)
```

Explanation: This query displays the main KPI values together so that they can be compared with the dashboard cards.

```
// Welcome to DAX query view! Learn more about DAX queries at https://aka.ms/dax-queries.
// Right-click on tables, columns, or measures in the data pane to access quick queries, or ask Copilot for help writing DAX.
EVALUATE
SUMMARIZECOLUMNS(
    "Total Sales", [Total Sales],
    "Total Profit", [Total Profit],
    "Total Orders", [Total Orders],
    "Total Customers", [Total Customers],
    "Profit Margin", [Profit Margin %]
)
```

Results | Result 1 of 1 | Copy

	[Total Sales]	[Total Profit]	[Total Orders]	[Total Customers]	[Profit Margin]
1	6404933.58	2601602.33	25164	17416	0.41

4.2 Query to Compare Sales by Year

```
EVALUATE
SUMMARIZECOLUMNS(
    'Calendar'[Year],
    "Total Sales", [Total Sales],
    "Total Profit", [Total Profit],
    "Total Orders", [Total Orders]
)
ORDER BY
    'Calendar'[Year]
```

Explanation: This query groups the results by year and displays Total Sales, Total Profit and Total Orders for each year.

It can help identify changes in sales and profitability across the available years.

```
EVALUATE
SUMMARIZECOLUMNS(
    'Calendar'[Year],
    "Total Sales", [Total Sales],
    "Total Cost", [Total Cost],
    "Total Profit", [Total Profit],
    "Profit Margin", [Profit Margin %]
)
ORDER BY
    'Calendar'[Year]
```

Results | Result 1 of 1 | Copy

	Calendar[Year]	[Total Sales]	[Total Cost]	[Total Profit]	[Profit Margin]
1	2020	6404933.58	3803331.25	2601602.33	0.41

```
EVALUATE
SUMMARIZECOLUMNS(
    'Calendar'[Year],
    'Calendar'[Month Number],
    'Calendar'[Month],
    "Total Sales", [Total Sales],
    "Previous Month Sales", [Previous Month Sales],
    "Sales Variance", [Sales Variance],
    "Sales Growth %", [Sales Growth %]
)
ORDER BY
    'Calendar'[Year],
    'Calendar'[Month Number]
```

Results | Result 1 of 1 | Copy

	Calendar[Year]	Calendar[Month Nu...	Calendar[Month]	[Total Sales]	[Previous Month Sal...	[Sales Variance]	[Sales Growth %]
1	2020	1	Jan	585312.65		585312.65	0
2	2020	2	Feb	532226.25	585312.65	-53086.4	-0.09
3	2020	3	Mar	643436.1	532226.25	111209.86	0.21
4	2020	4	Apr	653364.04	643436.1	9927.93	0.02
5	2020	5	May	659325.9	653364.04	5961.86	0.01
6	2020	6	Jun	669988.67	659325.9	10662.77	0.02
7	2020	7	Jul	486115.01	669988.67	-183873.66	-0.27
8	2020	8	Aug	536452.82	486115.01	50337.81	0.1

5. Dashboard Screenshot and Explanation

The dashboard was developed in Power BI Desktop to bring the main sales performance indicators and visual comparisons onto one page.

The design uses a blue AdventureWorks header, KPI cards and multiple visualizations. Slicers are positioned in the header area so that the report remains organized and does not contain duplicate slicers at the bottom.

5.1 KPI Cards

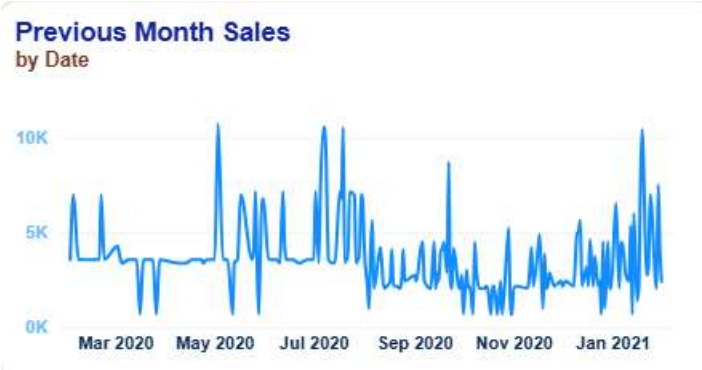
The dashboard contains five KPI cards.

KPI	Purpose
Total Sales	Shows the overall sales value for the selected data.
Total Profit	Shows the difference between sales and product cost.
Total Customers	Shows the number of unique customers.
Profit Margin %	Shows profit as a percentage of sales.
Average Order Quantity	Shows the average quantity per order or the average order quantity measure used in the model.

5.2 Line Chart — Total Sales and Previous Month Sales by Date

The line chart displays sales over time and compares current sales with previous-month sales.

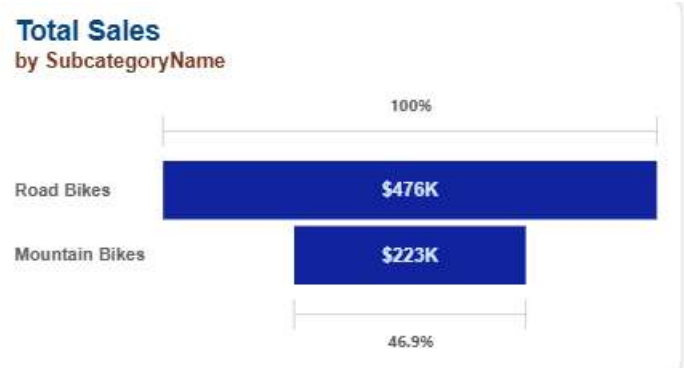
It is useful for identifying sales trends and changes between periods. The two lines help compare the current sales figures with the corresponding previous-period values.



5.3 Bar Chart — Total Sales by Subcategory

The bar chart compares sales across product subcategories.

It makes it easier to identify which subcategories contribute more to total sales. Longer bars represent higher sales values.



5.4 Donut Chart — Total Sales by Continent

The donut chart displays the distribution of sales across continents.

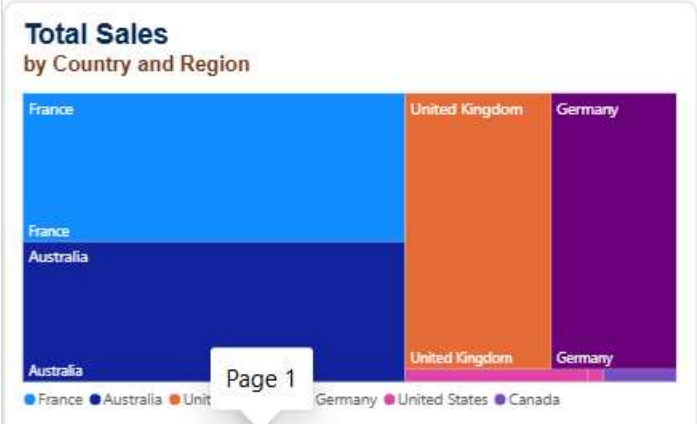
The size of each segment represents its contribution to the total. This provides a quick comparison of geographical sales performance.



5.5 Treemap — Total Sales by Country and Region

The treemap uses differently sized rectangles to represent sales across countries and regions.

Larger rectangles represent a greater sales contribution, making it easier to identify important geographical areas.



5.6 Scatter Plot — Total Sales and Total Profit

The scatter plot compares Total Sales and Total Profit for the selected grouping, such as country or continent.

It helps identify groups with high sales and high profit, as well as groups where sales and profit do not increase at the same rate.



5.7 Matrix — Sales and Profit Analysis

The matrix presents sales and profit values in rows and columns.

For example, region and product subcategory can be used to compare Total Sales, Total Profit and Profit Margin. This provides more detailed information than a single KPI card.

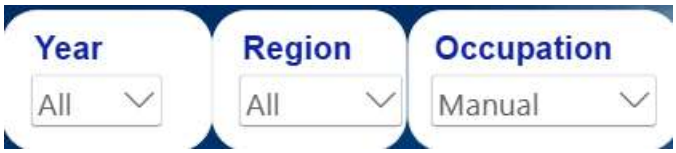
Region	Australia		Canada		France
SubcategoryName	Total Sales	Total Profit	Total Sales	Total Profit	Total Sales
Mountain Bikes	$82,556	$37,156.31	$3,400	$1,487.84	$51,123
Road Bikes	$1,15,897	$45,395.59			$1,58,988
Total	$1,98,453	$82,551.90	$3,400	$1,487.84	$2,10,111

5.8 Slicers and Interactivity

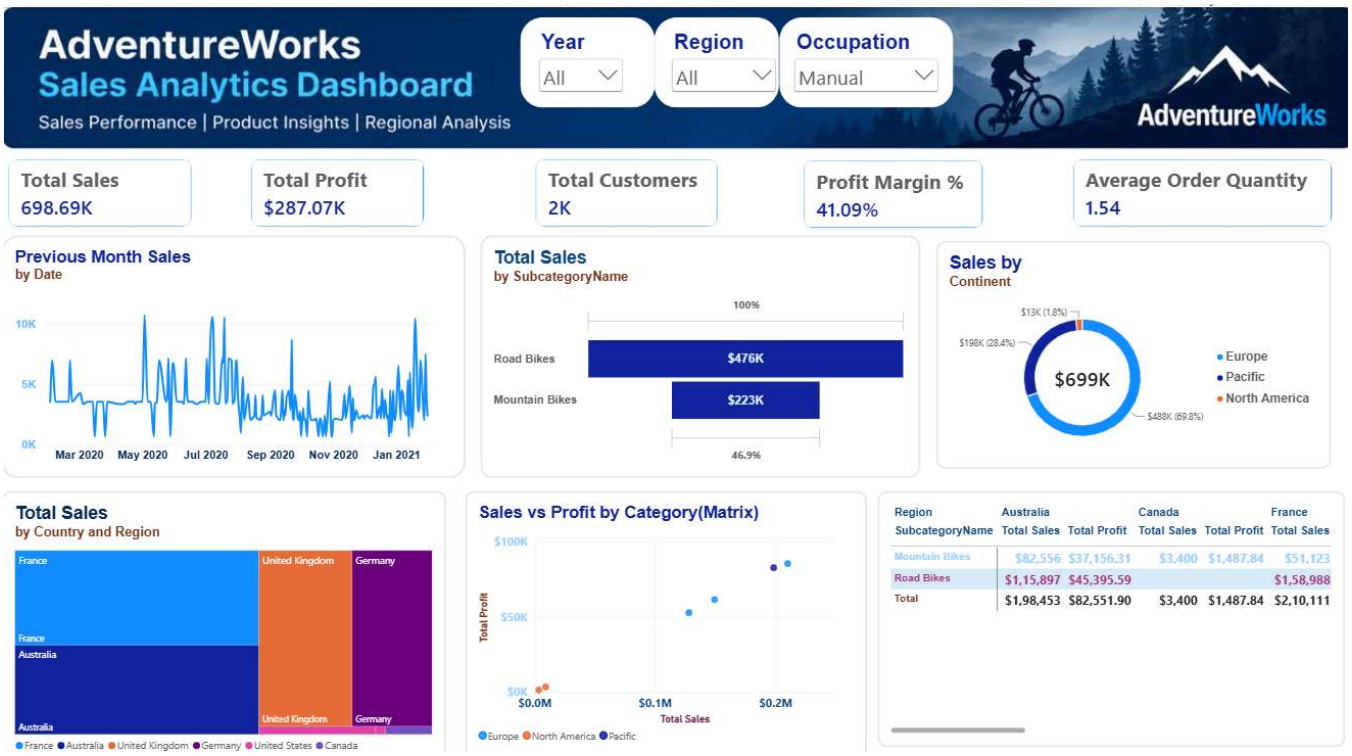
The dashboard includes slicers for Year, Region and Occupation, as shown in the current report design.

The slicers allow users to select a particular year, geographical region or occupation group and examine the corresponding results.

When a slicer selection changes, the KPI cards and visualizations should update according to the model relationships and visual interaction settings.



Final Dashboard Output



9. Conclusion

This practical helped me learn data preparation, DAX calculations, time-based analysis and dashboard creation in Power BI. The KPI cards, charts, matrix and slicers made it easier to analyze sales and profit from different perspectives. Overall, I gained practical experience in turning raw data into a useful and interactive dashboard.